

Implementation of a Multi-Router Network Using OSPF (Single Area) with VLANs, DHCP, DNS, and Web Server

Objectives

- To design a network consisting of three routers connected using OSPF in a single area (Area 0).
- To practice efficient subnetting (VLSM).
- To configure VLANs and trunking on switches.
- To implement DHCP with excluded IP ranges.
- To configure Web and DNS servers.
- To troubleshoot OSPF adjacency issues and routing failures.

Theory

OSPF (Open Shortest Path First)

OSPF (Open Shortest Path First) is a link-state routing protocol designed for IP networks to efficiently determine the best path for data. Unlike distance-vector protocols, OSPF builds a complete map of the network topology using Link-State Advertisements (LSAs) to all routers in the area. This allows routers to calculate the shortest path to each network using Dijkstra's Shortest Path First (SPF) algorithm.

Key Features:

- Classless routing protocol (supports VLSM and CIDR)
- Rapid convergence in case of network changes
- Cost metric based on bandwidth
- Supports hierarchical design using Areas to reduce routing overhead
- Uses multicast addresses 224.0.0.5 (All OSPF Routers) and 224.0.0.6 (All OSPF Designated Routers)

Subnetting (VLSM)

Variable Length Subnet Masking (VLSM) is a technique to divide an IP network into subnets of different sizes. VLSM allows efficient utilization of IP addresses, avoiding waste in networks where the number of hosts varies across subnets. Each subnet can have a custom subnet mask based on the number of required hosts.

Example:

Network	Required Hosts	Subnet Mask	Network Address
VLAN 10	50 Hosts	/26	192.168.10.0/26
VLAN 20	25 Hosts	/27	192.168.10.64/27
VLAN 30	25 Hosts	/27	192.168.10.96/27
VLAN 40	10 Hosts	/28	192.168.10.144/28

VLAN and Trunking

A Virtual LAN (VLAN) is a logical segmentation of a switch into multiple broadcast domains. VLANs improve network efficiency by isolating traffic, increasing security, and reducing unnecessary broadcast traffic.

Trunking allows multiple VLANs to share a single physical link between switches or between a switch and a router. IEEE 802.1Q encapsulation is commonly used to add VLAN tags to Ethernet frames, allowing switches and routers to identify which VLAN the traffic belongs to.

Benefits:

- Traffic segmentation and improved security
- Simplified network management
- Better utilization of resources

DHCP(Dynamic Host Configuration Protocol)

DHCP is a network protocol that automatically provides IP addresses, subnet masks, default gateways, and DNS server addresses to clients. This automation reduces administrative workload and ensures consistent configuration.

DHCP Process (DORA):

1. **Discover** – Client broadcasts a DHCPDISCOVER message.
2. **Offer** – DHCP server responds with a DHCPOFFER.
3. **Request** – Client requests the offered IP address.
4. **Acknowledge** – DHCP server sends a DHCPACK confirming the lease.

Excluded IP Ranges: Certain IP addresses (like gateways or server IPs) are excluded from DHCP to avoid conflicts.

DNS and Web Server

DNS (Domain Name System) converts human-readable domain names(e.g.,www.company.com) into IP addresses, allowing clients to locate resources on the network easily.

Web Servers use HTTP (Hypertext Transfer Protocol) to deliver web pages to clients. Setting up a web server along with DNS ensures that users can access resources using a domain name rather than an IP address, making the network more user-friendly.

Integration in Lab:

- DNS resolves the web server's domain name to its IP address.
- Web server hosts a website accessible from all VLANs, verifying proper routing and inter-VLAN communication.

Network Topology

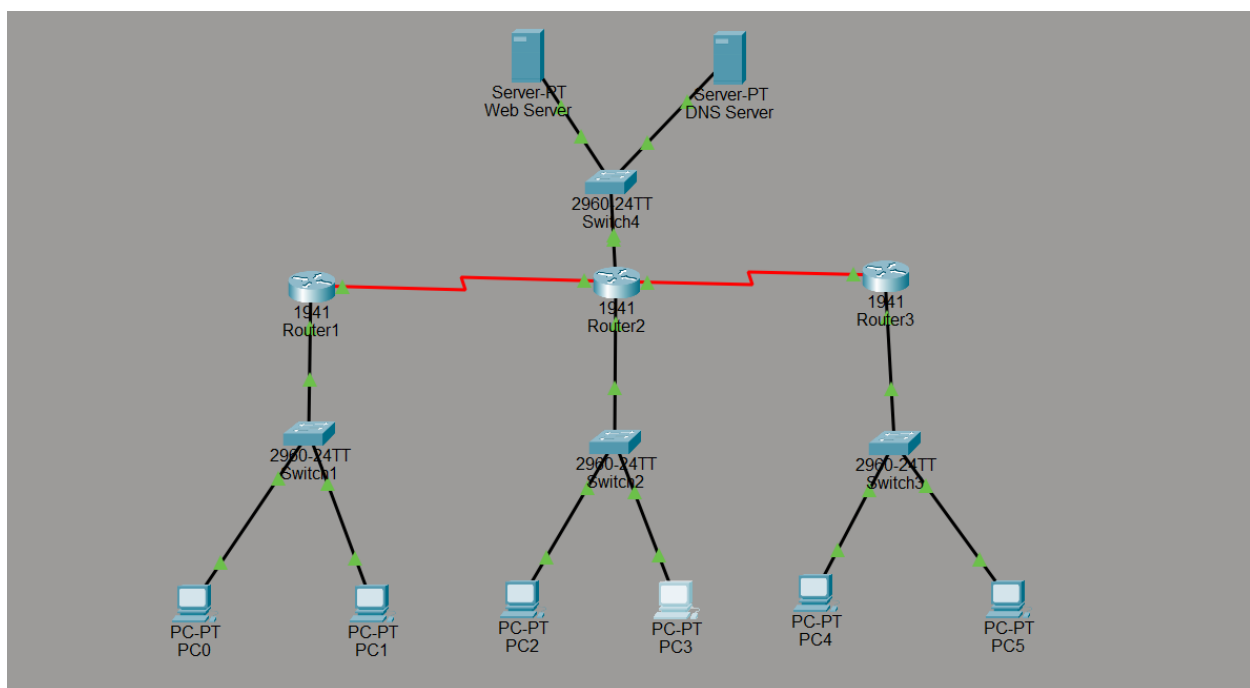


Fig: Network with 3 routers using OSPF in a single area

This topology illustrates a hub-and-spoke network connecting a central data center to two branch offices. Router2 acts as the central hub, providing access to shared resources like the Web and DNS servers via Switch4. The branch offices, represented by Router1 and Router3, connect to this hub over wide-area network links. Locally, each branch uses switches to connect user workstations to their respective routers. This design allows for centralized management of resources while connecting geographically dispersed sites.

IP Addressing Scheme

The network uses the 192.168.10.0/24 subnet, efficiently managed using Variable Length Subnet Masking (VLSM).

Network	VLAN	Subnet	Gateway
HR	10	192.168.10.0/26	192.168.10.1
Sales	20	192.168.10.64/27	192.168.10.65
IT	30	192.168.10.96/27	192.168.10.97
Servers	40	192.168.10.144/28	192.168.10.145

Configuration Steps

1. Router Configuration

Routers were configured with sub-interfaces using 802.1Q encapsulation for inter-VLAN routing and OSPF for dynamic routing between sites.

R1 Configuration(HR)

```
enable
conf t
hostname R1

interface g0/0
ip address 192.168.10.1 255.255.255.192
no shutdown

interface s0/0/0
ip address 192.168.10.129 255.255.255.252
no shutdown

router ospf 1
network 192.168.10.0 0.0.0.63 area 0
network 192.168.10.128 0.0.0.3 area 0

ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp pool HR
network 192.168.10.0 255.255.255.192
default-router 192.168.10.1
dns-server 192.168.10.151
```

R2 Configuration(SALES)

```
enable
conf t
hostname R2

interface g0/0.20
 encapsulation dot1Q 20
 ip address 192.168.10.65 255.255.255.224

interface g0/1.40
 encapsulation dot1Q 40
 ip address 192.168.10.145 255.255.255.240

interface s0/0/0
 ip address 192.168.10.130 255.255.255.252
 no shutdown
interface s0/0/1
 ip address 192.168.10.133 255.255.255.252
 no shutdown

router ospf 1
 passive-interface g0/0.20
 passive-interface g0/1.40
 network 192.168.10.64 0.0.0.31 area 0
 network 192.168.10.128 0.0.0.3 area 0
 network 192.168.10.132 0.0.0.3 area 0
 network 192.168.10.144 0.0.0.15 area 0

ip dhcp excluded-address 192.168.10.65 192.168.10.70
ip dhcp pool SALES
 network 192.168.10.64 255.255.255.224
 default-router 192.168.10.65
 dns-server 192.168.10.151
```

R3 Configuration(IT)

```
enable
conf t
hostname R3

interface g0/0.30
 encapsulation dot1Q 30
 ip address 192.168.10.97 255.255.255.224

interface s0/0/0
 ip address 192.168.10.134 255.255.255.252
 no shutdown
```

```
router ospf 1
passive-interface g0/0.30
network 192.168.10.96 0.0.0.31 area 0
network 192.168.10.132 0.0.0.3 area 0

ip dhcp excluded-address 192.168.10.97 192.168.10.100
ip dhcp pool IT
network 192.168.10.96 255.255.255.224
default-router 192.168.10.97
dns-server 192.168.10.151
```

2. Switch Configuration

Switch ports connecting to routers were configured as trunks to allow traffic for multiple VLANs to pass through.

Switch 1 (HR Site)

```
vlan 10
name HR

interface range f0/1-2
switchport mode access
switchport access vlan 10

interface fa0/3
switchport mode trunk
```

Switch 2 (Sales Site)

```
vlan 10
name HR

interface range f0/1-2
switchport mode access
switchport access vlan 10

interface fa0/3
switchport mode trunk
```

Switch 3 (IT Site)

```
vlan 30
name IT

interface range f0/1-2
switchport mode access
switchport access vlan 30
interface fa0/3
switchport mode trunk
```

Switch 4(Server Site)

vlan 40

name SERVERS

interface range fa0/1-2

switchport mode access

switchport access vlan 40

interface fa0/24

switchport mode trunk

3. DNS and Web Server Setup

Web Server

IP: 192.168.10.150

Mask: 255.255.255.240

Gateway: 192.168.10.145

DNS: 192.168.10.151

Turn HTTP ON.

DNS Server

IP: 192.168.10.151

Mask: 255.255.255.240

Gateway: 192.168.10.145

Add record: company.local → 192.168.10.150

4. Set PCs to DHCP

Go to each PC:

Desktop → IP Configuration → DHCP

5. Troubleshooting

OSPF Adjacency Issues

- Check: Area ID, Hello/Dead timers, subnet, passive interface
- Commands:

show ip ospf neighbor

debug ip ospf adj

Routing Failures

- Check: Network statements, interface status, wildcard masks
- Commands:

show ip route

show ip protocols

VLAN Issues

- Check VLAN existence, port assignment, trunk allowed VLANs
- Commands:
 - show vlan brief
 - show interfaces trunk

DHCP Issues

- Check excluded ranges, default gateway, DHCP pool
- Commands:
 - show ip dhcp binding
 - show ip dhcp pool

Verification and Testing

Connectivity Testing

Connectivity was verified by pinging from the HR PC, Sales PC and IT PC to the Web Server (192.168.10.150). Successful replies confirmed that inter-VLAN routing, trunking, and OSPF routing are functioning correctly.

```
C:\>ping 192.168.10.150

Pinging 192.168.10.150 with 32 bytes of data:

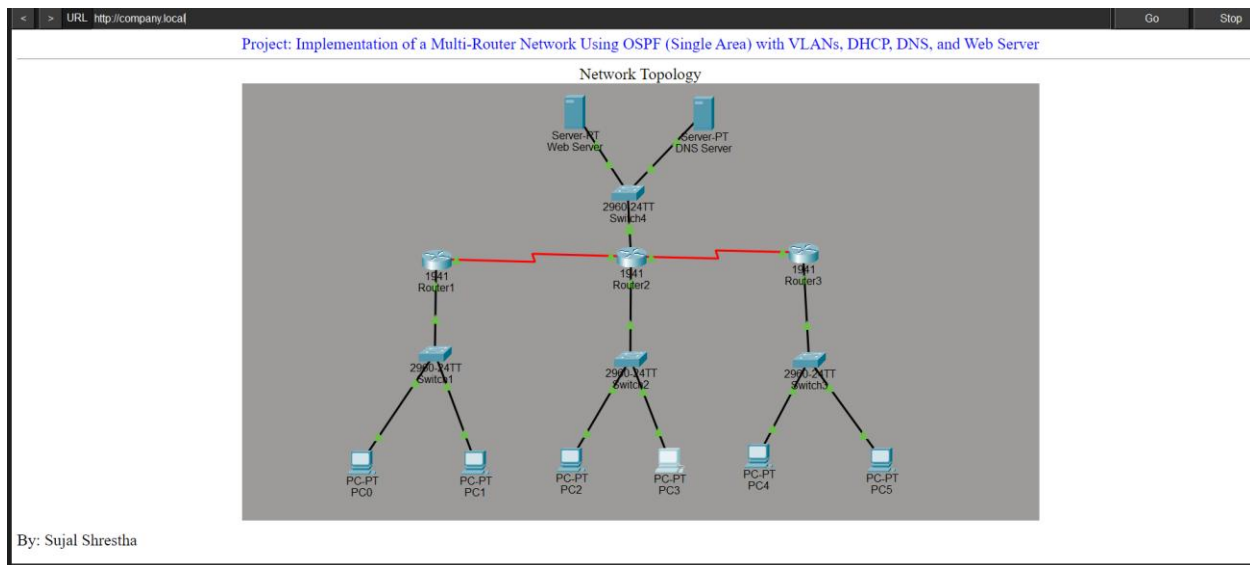
Reply from 192.168.10.150: bytes=32 time=1ms TTL=126
Reply from 192.168.10.150: bytes=32 time=1ms TTL=126
Reply from 192.168.10.150: bytes=32 time=10ms TTL=126
Reply from 192.168.10.150: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.10.150:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 5ms
```

Fig: Ping test from PC0

DNS and HTTP Verification

Testing was finalized by accessing the web server via the browser using the domain name <http://company.local>. The successful loading of the webpage confirms the coordination of DHCP, DNS, and HTTP services across the expanded topology.



Discussion

The primary challenge faced during this project was achieving connectivity between the clients and the Web Server after introducing **Switch 4** to manage the server network.

Issue Identification

Initial ping tests from the client PCs to the Web Server failed. Investigation revealed two critical Layer 2 and Layer 3 issues:

1. **Missing Inter-VLAN Routing:** Initially, routers used physical interfaces for VLAN gateways, which prevented handling traffic tagged for specific VLANs from the switches.
2. **Trunking Misconfiguration:** The link between **Switch 4** and **Router 2** was not properly configured as a trunk port, preventing VLAN 40 traffic from reaching the router.
3. **Spanning Tree Blocking:** **Switch 4** was incorrectly electing a root bridge or experiencing a Spanning Tree Protocol (STP) loop, causing the port connecting to the router to be placed in a BLOCKING state.

Solutions Implemented

To resolve these issues, the following actions were taken:

1. **Implemented "Router-on-a-Stick":** Configured sub-interfaces (g0/0.20, g0/0.30, etc.) on **R2** and **R3** with encapsulation dot1Q to correctly handle VLAN tags.
2. **Trunk Configuration:** The interface on **Switch 4** connecting to **Router 2** was manually configured with switchport mode trunk to allow VLAN 40 traffic to pass.

3. **STP Troubleshooting:** To resolve the blocking issue, spanning-tree portfast trunk was configured on the switch port connected to the router. This allowed the port to transition immediately to a forwarding state, resolving the connectivity failure.

Conclusion

This report demonstrated the practical implementation of a multi-router OSPF network in a single area with efficient subnetting using VLSM. VLAN segmentation and trunking improved network organization and security, while DHCP automation simplified IP management and DNS provided application-layer functionality. Troubleshooting steps enhanced understanding of routing and configuration issues.

Specifically, the experiment successfully integrated routing, switching, and network services into a complete enterprise-style network design. Furthermore, the project highlighted the critical need for meticulous Layer 2 configuration, evidenced by the successful resolution of Spanning Tree Protocol blocking issues on Switch 4 and the implementation of necessary trunking on Router 2. Ultimately, these efforts ensured robust, end-to-end connectivity between all client VLANs and the centralized server resources.